Positive-density nil-Bohr recurrent sets are 2-large

Produced by ChatGPT-6 pro, prompted by Sohail Farhangi

4 November 2026

Abstract

We prove that a set of positive integers which is nil-Bohr recurrent at every finite order and has positive upper Banach density is 2-large. Equivalently, every nil-Bohr recurrent set admitting a binary obstruction to arithmetic progressions of some fixed length has zero upper Banach density. For a fixed length H, recurrence for $(H-1)$-step nilsystems suffices. The main auxiliary result is that pointwise recurrence for finite-dimensional nilsystems implies measurable multiple recurrence in those systems. We prove this auxiliary result using the closedness theorems of Mozes and Shah for homogeneous measures, including their closedness statement with a prescribed point in the support. A conditional-expectation argument and the scalar characteristic-factor theorem of Bergelson, Host, and Kra then give the density conclusion. All structural inputs are stated explicitly, and the argument does not assume pointwise decay of nilsystem correlation sequences.

1 Definitions and main result

We write $\mathbb{N} = \{1, 2, \ldots\}$. An arithmetic progression of length H is a set $\{a, a+d, \ldots, a+(H-1)d\}$ with $a, d \in \mathbb{N}$. A set $D \subseteq \mathbb{N}$ is 2-*large* if every map $c : \mathbb{N} \to \{0,1\}$ admits, for every $H \geq 2$, a monochromatic progression of length H with common difference in D. The upper Banach density of D is

$$d^*(D) = \lim_{N\to\infty} \sup_{M\geq 0} \frac{|D \cap \{M+1, \ldots, M+N\}|}{N}. \tag{1.1}$$

The limit exists by subadditivity. If the natural density $d(D) = \lim_{N\to\infty} |D \cap \{1, \ldots, N\}|/N$ exists, then $d(D) \leq d^*(D)$.

A finite-dimensional s-step nilsystem is a translation $S : y \mapsto ay$ on a compact nilmanifold $Y = G/\Gamma$, where G is a nilpotent Lie group of class at most s and Γ is a discrete cocompact subgroup. We allow disconnected G and disconnected Y. The Haar probability measure on Y is denoted by ν. For $y \in Y$ and an open neighborhood V of y, set

$$N_S(y, V) = \{n \in \mathbb{N} : S^n y \in V\}. \tag{1.2}$$

A set containing such a return-time set is called a *nil-s-Bohr neighborhood of zero*. A set $D \subseteq \mathbb{N}$ is *nil-s-Bohr recurrent* if it meets every such neighborhood. It is *nil-Bohr recurrent at every finite order* if it is nil-s-Bohr recurrent for every $s \geq 1$. Finite products and powers preserve the step bound. Finite cyclic rotations are included as 1-step nilsystems. Equivalently, one may restrict the definition to minimal nilsystems by passing to point orbit closures.

Theorem 1.1. *Let $H \geq 2$ and let $D \subseteq \mathbb{N}$ be nil-$(H-1)$-Bohr recurrent. Suppose there is a coloring $c : \mathbb{N} \to \{0,1\}$ such that*

$$c(a), c(a+d), \ldots, c(a+(H-1)d) \quad \text{are not all equal} \tag{1.3}$$

for every $a \in \mathbb{N}$ and $d \in D$. Then $d^*(D) = 0$.

Corollary 1.2. *If D is nil-Bohr recurrent at every finite order and $d^*(D) > 0$, then D is 2-large. In particular, there is no nil-Bohr recurrent set of positive natural density which is not 2-large. Indeed, every nil-Bohr recurrent set which is not 2-large has natural density zero.*

Proof. Failure of 2-largeness supplies an H and a coloring satisfying (1.3). Theorem 1.1 then gives $d^*(D) = 0$. Equation (1.1) implies $|D \cap \{1, \ldots, N\}|/N \to 0$. □

The proof has two parts. First we obtain measurable multiple recurrence inside a finite-dimensional nilsystem from pointwise nilsystem recurrence. Then we apply this to the finite nilsystem factors of an ergodic symbolic model of the coloring. The resulting lower bounds for conditional color probabilities force the obstructed differences into a set of zero upper Banach density. The external inputs are the homogeneous-measure theorems of Mozes–Shah [3], the Host–Kra structure theorem [2], and the scalar correlation estimate in Bergelson–Host–Kra [1].

2 Two elementary lemmas about measures

Lemma 2.1 (Products and fixed marginals)**.** *Let Y be a compact metric space with Borel probability measure ν. Let σ_ℓ, σ be probability measures on Y^H such that $\sigma_\ell \to \sigma$ weakly and every coordinate marginal of every σ_ℓ is ν. Then every coordinate marginal of σ is ν, and for every Borel set $B \subseteq Y$,*

$$\sigma_\ell(B^H) \longrightarrow \sigma(B^H). \tag{2.1}$$

Proof. The assertion about the marginals follows by testing continuous functions of one coordinate. Given $\varepsilon > 0$, choose $f \in C(Y)$ with $0 \leq f \leq 1$ and $\|f - \mathbf{1}_B\|_{L^1(\nu)} < \varepsilon$. For any probability measure τ on Y^H with these marginals, the telescoping product inequality gives

$$\begin{aligned} \left| \tau(B^H) - \int_{Y^H} \prod_{j=0}^{H-1} f(y_j)\, d\tau \right| &\leq \sum_{j=0}^{H-1} \int_{Y^H} |\mathbf{1}_B(y_j) - f(y_j)|\, d\tau \\ &\leq H\varepsilon. \end{aligned} \tag{2.2}$$

Apply this to σ_ℓ and σ. The integrals of the continuous product converge, so $\limsup_\ell |\sigma_\ell(B^H) - \sigma(B^H)| \leq 2H\varepsilon$. Let $\varepsilon \downarrow 0$. Completed measurable sets can be replaced by Borel representatives, since all coordinate marginals are ν. □

Lemma 2.2 (A smooth measure containing the diagonal)**.** *Let Y be a compact smooth manifold with a smooth strictly positive probability density ν. Let $W \subseteq Y^H$ be a closed smooth embedded submanifold containing*

$$\Delta Y = \{(y, \ldots, y) : y \in Y\}. \tag{2.3}$$

Let σ be a probability measure on W whose density in smooth local coordinates is smooth and strictly positive. Then

$$\nu(B) > 0 \quad \Longrightarrow \quad \sigma(B^H) > 0 \tag{2.4}$$

for every measurable $B \subseteq Y$.

Proof. Choose a Lebesgue density point y of B and put $z = (y, \dots, y) \in W$. Write $r = \dim Y$ and $m = \dim W$. The zero-dimensional case is immediate, so assume $r > 0$. For the coordinate projection $p_j : W \to Y$, the differential at z is surjective: $T_z W$ contains $T_z(\Delta Y)$, and p_j is the identity on the diagonal tangent space. Thus every p_j is a submersion near z.

Fix smooth coordinate balls V_ρ in W centered at z. For small $\rho > 0$, smoothness and positivity of the density give

$$\sigma(V_\rho) \geq c\rho^m \tag{2.5}$$

for a constant $c > 0$. The submersion theorem gives local coordinates $(u, v) \in \mathbb{R}^r \times \mathbb{R}^{m-r}$ in which $p_j(u, v) = u$. On a sufficiently small fixed neighborhood, the coordinate changes and the measure densities have bounded Jacobians. The image of V_ρ in these coordinates lies in a box of side at most $C\rho$. Integrating first in v, and comparing the u-coordinate volume with ν, yields

$$\sigma\big(V_\rho \cap p_j^{-1}(Y \setminus B)\big) \leq C\rho^{m-r}\, \nu\big(B_Y(y, C\rho) \setminus B\big). \tag{2.6}$$

Enlarge C to work for all the finitely many coordinate projections. Since y is a density point, $\nu(B_Y(y, C\rho) \setminus B) = o(\rho^r)$. Summing (2.6) over j gives

$$\sigma(V_\rho \setminus B^H) = o(\rho^m). \tag{2.7}$$

Together with (2.5), this implies $\sigma(V_\rho \cap B^H) > 0$ for all sufficiently small ρ. □

3 The homogeneous-measure input

Let L be a connected Lie group, let $\Lambda \leq L$ be discrete, and let $Z = L/\Lambda$. For a probability measure σ on Z, let $\Lambda(\sigma)$ denote its invariance group under the action of L. Following [3], let $\mathcal{Q}(Z)$ be the class of probability measures for which the subgroup generated by the Ad-unipotent one-parameter subgroups contained in $\Lambda(\sigma)$ acts ergodically. For $z \in Z$, set

$$\mathcal{Q}_z(Z) = \{\sigma \in \mathcal{Q}(Z) : z \in \operatorname{supp} \sigma\}. \tag{3.1}$$

Theorem 3.1 (Mozes–Shah)**.** *The classes $\mathcal{Q}(Z)$ and $\mathcal{Q}_z(Z)$ are weakly closed. Every measure in $\mathcal{Q}(Z)$ is homogeneous: its support is a closed subgroup orbit, and it is the invariant probability measure on that orbit.*

The closedness statements are Corollaries 1.1 and 1.2 in the author version of [3] linked in the bibliography. The homogeneity assertion is Ratner's measure-classification theorem in the form recalled there. A closed homogeneous orbit in this setting is a smooth embedded submanifold; see also [3, Lemma 2.1]. Its invariant measure is smooth and strictly positive in local coordinates. We will apply the theorem only to compact homogeneous spaces of connected nilpotent Lie groups. In such a group, every one-parameter subgroup is Ad-unipotent.

Remark 3.2. The closedness of $\mathcal{Q}_z(Z)$ is stronger than ordinary weak compactness. For arbitrary measures, points in the supports can converge to a point outside the support of the limiting measure. The argument below uses the prescribed-point closedness statement directly.

4 Measurable recurrence in finite-dimensional nilsystems

Theorem 4.1. *Let $s \geq 1$, and let $D \subseteq \mathbb{N}$ be nil-s-Bohr recurrent. Let $(Y = G/\Gamma, \nu, S)$ be an s-step nilsystem with its Haar probability measure, let $\nu(B) > 0$, and let $H \geq 2$. There is a $d \in D$ such that*

$$\nu\left(\bigcap_{j=0}^{H-1} S^{-jd} B\right) > 0. \tag{4.1}$$

Proof. We first suppose that Y is connected. The presenting group G may still be disconnected. Write $S(y) = ay$ and let G^0 be the identity component of G. The G^0-orbits in Y are open, since Γ is discrete. Since Y is connected, G^0 acts transitively, so Y is a compact homogeneous space of G^0 with a discrete stabilizer.

Step 1: simultaneous returns on a dense set. Choose a countable dense set $\{y_i : i \geq 1\}$ in Y. For each $\ell \geq 1$, apply nil-s-Bohr recurrence to the finite product of copies of (Y, S^j), indexed by $1 \leq i \leq \ell$ and $1 \leq j < H$, with base points y_i. Products and powers remain s-step nilsystems. We obtain $d_\ell \in D$ such that

$$\operatorname{dist}(S^{jd_\ell} y_i, y_i) < \frac{1}{\ell} \qquad (1 \leq i \leq \ell,\ 1 \leq j < H). \tag{4.2}$$

In particular, for each fixed i and j, $S^{jd_\ell} y_i \to y_i$.

Step 2: homogeneous graph measures. Define

$$\sigma_\ell = (\Phi_\ell)_* \nu, \qquad \Phi_\ell(y) = (y, S^{d_\ell} y, \dots, S^{(H-1)d_\ell} y). \tag{4.3}$$

Every coordinate marginal of σ_ℓ is ν. The graph $\Phi_\ell(Y)$ is a compact orbit of the subgroup

$$L_\ell = \left\{ \left(a^{jd_\ell} g a^{-jd_\ell}\right)_{j=0}^{H-1} : g \in G^0 \right\} \ \leq\ (G^0)^H. \tag{4.4}$$

Here the zeroth coordinate is g. The inclusion follows because G^0 is normal in G. For $g \in G^0$, the corresponding element of L_ℓ sends $\Phi_\ell(y)$ to $\Phi_\ell(gy)$. Thus L_ℓ preserves σ_ℓ and acts transitively on its support. It is connected and generated by Ad-unipotent one-parameter subgroups of the connected nilpotent ambient group $(G^0)^H$. It follows that $\sigma_\ell \in \mathcal{Q}(Y^H)$.

Pass to a weakly convergent subsequence, still denoted σ_ℓ, with limit σ. Theorem 3.1 implies that σ is homogeneous. Let $W = \operatorname{supp} \sigma$.

Step 3: the entire diagonal belongs to W. Fix i and put

$$z_\ell = \Phi_\ell(y_i), \qquad z = (y_i, \dots, y_i). \tag{4.5}$$

Then $z_\ell \in \operatorname{supp} \sigma_\ell$ and $z_\ell \to z$. The orbit map of $(G^0)^H$ is a smooth submersion and admits a local section near z. Consequently there are $b_\ell \to e$ in $(G^0)^H$ such that $b_\ell z_\ell = z$. Translation preserves the class $\mathcal{Q}$, so

$$(b_\ell)_* \sigma_\ell \in \mathcal{Q}_z(Y^H), \qquad (b_\ell)_* \sigma_\ell \longrightarrow \sigma. \tag{4.6}$$

The prescribed-point closedness in Theorem 3.1 gives $z \in W$. This holds for every y_i. By density and closedness,

$$\Delta Y \subseteq W. \tag{4.7}$$

Step 4: recover a positive measurable intersection. The measure σ is smooth and strictly positive on the closed homogeneous submanifold W. Lemma 2.2, together with (4.7), gives $\sigma(B^H) > 0$. All coordinate marginals are ν, so Lemma 2.1 gives

$$\nu\left(\bigcap_{j=0}^{H-1} S^{-jd_\ell} B\right) = \sigma_\ell(B^H) \longrightarrow \sigma(B^H) > 0. \tag{4.8}$$

This proves (4.1) for some $d_\ell \in D$.

Step 5: disconnected nilmanifolds. A compact manifold has only finitely many connected components. Choose a component Y_0 with $\nu(B \cap Y_0) > 0$ and an integer $q \geq 1$ such that $S^q Y_0 = Y_0$. Let ν_0 be normalized Haar measure on Y_0, and write $B_0 = B \cap Y_0$. In Step 1, use a dense set in Y_0, but take the recurrence tests in the original systems (Y, S^j). Adjoin the cyclic rotation on $\mathbb{Z}/q\mathbb{Z}$, based at 0 with neighborhood $\{0\}$. This yields $d_\ell \in D \cap q\mathbb{N}$ satisfying (4.2). Every S^{jd_ℓ} preserves Y_0 and ν_0.

Now form the graph measures on Y_0^H using ν_0. The connected group G^0 acts transitively on Y_0, and the subgroups in (4.4) lie in $(G^0)^H$ and have the same transitivity and invariance properties. Steps 2–4 apply verbatim with (Y_0, ν_0, B_0) in place of (Y, ν, B). The positive ν_0-intersection gives a positive ν-intersection. This reduction uses no recurrence claim about a divided set D/q and does not require S^q to be a translation by an element of G^0. □

Corollary 4.2. *Let (Y, ν, S) be an s-step nilsystem, let $\nu(B) > 0$, and let $H \geq 2$. Then*

$$R_H^\nu(B) = \left\{ n \in \mathbb{N} : \nu\left(\bigcap_{j=0}^{H-1} S^{-jn} B \right) > 0 \right\} \tag{4.9}$$

contains a nil-s-Bohr neighborhood of zero.

Proof. If $R_H^\nu(B)$ contained no such neighborhood, its complement would meet every nil-s-Bohr neighborhood and hence would be nil-s-Bohr recurrent. Theorem 4.1 applied to that complement gives a contradiction. □

5 The scalar characteristic-factor input

For an ergodic invertible probability-preserving system (X, μ, T) on a standard probability space, write $\mathcal{Z}_k$ for its Host–Kra factor of order k. We use the following two structural facts.

Theorem 5.1 (Host–Kra and Bergelson–Host–Kra)**.** *Let (X, μ, T) be an ergodic invertible probability-preserving system on a standard probability space, and let $k \geq 1$.*

(i) *There are increasing invariant sub-σ-algebras $\mathcal{Y}_m$ generating $\mathcal{Z}_k$ modulo null sets, such that each corresponding factor is an ergodic finite-dimensional k-step nilsystem.*

(ii) *For a bounded real measurable function f, put $g = \mathbb{E}(f \mid \mathcal{Z}_k)$ and*

$$I_f(k, n) = \int_X \prod_{j=0}^{k} f(T^{jn} x)\, d\mu(x). \tag{5.1}$$

Then $e(n) = I_f(k, n) - I_g(k, n)$ satisfies

$$\lim_{N \to \infty} \sup_{M \in \mathbb{Z}} \frac{1}{N} \sum_{n=M}^{M+N-1} |e(n)| = 0. \tag{5.2}$$

Part (i) is the Host–Kra structure theorem [2, Theorem 10.1], applied to $\mathcal{Z}_k$. Part (ii) is [1, Corollary 4.6]. The terminology *convergence to zero in uniform density* refers to (5.2); see [1, Definition 1.7]. In particular, every fixed positive-threshold level set of $|e(n)|$ has zero upper Banach density. The index needed for H sites is $k = H - 1$.

6 A bound for the conditional color probabilities

Lemma 6.1. *Let (X, μ, T) be an invertible probability-preserving system, let $H \geq 2$, and let $A \subseteq X$ be measurable. Suppose that for every $d \in D$,*

$$\mu\left(\bigcap_{j=0}^{H-1} T^{-jd}A\right) = \mu\left(\bigcap_{j=0}^{H-1} T^{-jd}A^c\right) = 0. \tag{6.1}$$

Let $\pi : (X, \mu, T) \to (Y, \nu, S)$ be an s-step nilsystem factor, and suppose D is nil-s-Bohr recurrent. Then $p = \mathbb{E}(\mathbf{1}_A \mid Y)$ satisfies

$$\frac{1}{H} \leq p \leq 1 - \frac{1}{H} \qquad \nu\text{-almost everywhere.} \tag{6.2}$$

Proof. Suppose first that $\nu(\{p < 1/H\}) > 0$. Choose $\varepsilon > 0$ such that

$$B = \left\{y : p(y) \leq \frac{1}{H} - \varepsilon\right\} \tag{6.3}$$

has positive measure. By Theorem 4.1, there is $d \in D$ such that

$$C = \bigcap_{j=0}^{H-1} S^{-jd}B \tag{6.4}$$

has positive measure. Conditional expectation commutes with the transformations of a factor, so

$$\mu(\pi^{-1}C \cap T^{-jd}A) = \int_C p(S^{jd}y)\, d\nu(y). \tag{6.5}$$

The union bound and (6.5) imply

$$\begin{aligned}\mu\left(\pi^{-1}C \cap \bigcup_{j=0}^{H-1} T^{-jd}A\right) &\leq \int_C \sum_{j=0}^{H-1} p(S^{jd}y)\, d\nu(y) \\ &\leq (1 - H\varepsilon)\nu(C).\end{aligned} \tag{6.6}$$

Since $\mu(\pi^{-1}C) = \nu(C)$, taking the complement inside $\pi^{-1}C$ gives

$$\mu\left(\pi^{-1}C \cap \bigcap_{j=0}^{H-1} T^{-jd}A^c\right) \geq H\varepsilon\nu(C) > 0. \tag{6.7}$$

This contradicts (6.1). Therefore $p \geq 1/H$ almost everywhere. Apply the same argument to A^c to obtain $1 - p \geq 1/H$. □

Remark 6.2. No uniform lower bound for $\nu(C)$ is needed. The union bound is taken inside $\pi^{-1}C$, so the positive lower bound is relative to $\nu(C)$ itself. Also, only finite-dimensional nilsystem factors occur in this lemma. There is no need to assert measurable recurrence for arbitrary sets in an entire inverse-limit nilsystem.

7 Proof of the main theorem

Proof of Theorem 1.1. Fix the coloring c in (1.3). Extend it arbitrarily to a point $x_* \in \{0,1\}^{\mathbb{Z}}$ and let T be the left shift, $(Tx)(n) = x(n+1)$. Define its forward limit set

$$X = \bigcap_{M \geq 1} \overline{\{T^m x_* : m \geq M\}}. \tag{7.1}$$

This is a nonempty compact T-invariant set, and T restricts to a homeomorphism of X. Every finite pattern in a point of X occurs arbitrarily far to the right in x_*, where x_* agrees with c. Consequently no point of X has a monochromatic H-term progression of coordinates with difference in D.

Choose an ergodic T-invariant Borel probability measure μ on X, and let

$$A = \{x \in X : x(0) = 0\}. \tag{7.2}$$

For every $d \in D$, both intersections in (6.1) are in fact empty, so their measures are zero.

Let $k = H - 1$ and use Theorem 5.1(i) to write $\mathcal{Z}_k$ as an increasing inverse limit of k-step nilsystem factors Y_m. Set

$$p_m = \mathbb{E}(\mathbf{1}_A \mid Y_m), \qquad g = \mathbb{E}(\mathbf{1}_A \mid \mathcal{Z}_k). \tag{7.3}$$

The hypothesis on D is precisely nil-k-Bohr recurrence. Lemma 6.1 therefore gives

$$\frac{1}{H} \le p_m \le 1 - \frac{1}{H} \qquad \text{almost everywhere for every } m. \tag{7.4}$$

By martingale convergence, the same bounds hold for g. Since T preserves μ, it follows for every $n \in \mathbb{Z}$ that

$$b(n) := \int_X \prod_{j=0}^{H-1} g(T^{jn}x)\, d\mu(x) \ge H^{-H}. \tag{7.5}$$

Put

$$a(n) = \int_X \prod_{j=0}^{H-1} \mathbf{1}_A(T^{jn}x)\, d\mu(x), \qquad e(n) = a(n) - b(n). \tag{7.6}$$

Theorem 5.1(ii) says that $e(n)$ tends to zero in uniform density. For $n \in D$, however, $a(n) = 0$. Equation (7.5) therefore implies

$$|e(n)| \ge H^{-H} \qquad (n \in D). \tag{7.7}$$

For every $M \ge 0$ and $N \ge 1$,

$$\frac{|D \cap \{M+1, \dots, M+N\}|}{N} \le H^H \frac{1}{N} \sum_{n=M+1}^{M+N} |e(n)|. \tag{7.8}$$

Take the supremum over M and let $N \to \infty$. Uniform-density convergence gives $d^*(D) = 0$. □

8 Consequences and scope

The argument also controls the full set of obstructed differences for the witnessing coloring. For $c : \mathbb{N} \to \{0, 1\}$, define

$$\mathrm{Bad}_H(c) = \{d \in \mathbb{N} : \text{there is no monochromatic } H\text{-term progression of difference } d\}. \tag{8.1}$$

Corollary 8.1. *If* $\mathrm{Bad}_H(c)$ *contains a nil-*$(H-1)$*-Bohr recurrent set, then* $d^*(\mathrm{Bad}_H(c)) = 0$. *Equivalently, for every binary coloring* c *and every* $H \ge 2$, *at least one of the following holds:*

(i) $d^*(\mathrm{Bad}_H(c)) = 0$;

(ii) *the good* H*-term differences contain a nil-*$(H-1)$*-Bohr neighborhood of zero.*

Proof. Nil-$(H-1)$-Bohr recurrence is upward closed. Thus, under the first hypothesis, $\mathrm{Bad}_H(c)$ is itself recurrent in that sense, and Theorem 1.1 applies to it. For the dichotomy, if $\mathrm{Bad}_H(c)$ is not nil-$(H-1)$-Bohr recurrent, it misses one of the defining neighborhoods, which is then contained in its complement. □

Relation to piecewise syndeticity. A set is piecewise syndetic if finitely many of its translates have thick union. Every piecewise syndetic set has positive upper Banach density, since upper Banach density is translation invariant and finitely subadditive, and a thick set has upper Banach density 1. Thus Corollary 1.2 includes the piecewise-syndetic obstruction and applies also to positive-density sets which are not piecewise syndetic. In particular, adjoining any set of positive upper Banach density to a set that is nil-Bohr recurrent at every finite order always produces a 2-large set.

Why two colors matter. Lemma 6.1 uses A and its entire complement as the two colors. It forces both conditional color probabilities to lie in $[1/H, 1-1/H]$. For a partition into three or more cells, bounds of the form $p_i \le 1-1/H$ do not give a positive lower bound for each p_i. The uniform lower bound (7.5) therefore comes from the binary hypothesis. No assertion of largeness for arbitrarily many colors is made here.

Logical dependencies. The measurable nilsystem recurrence theorem uses the two elementary measure lemmas and the two closedness statements in Mozes–Shah. The main theorem then uses that recurrence theorem, a conditional union bound, the Host–Kra inverse-limit structure, and the Bergelson–Host–Kra uniform-density estimate. In particular, no step upgrades a general nilsequence-plus-null decomposition to a pointwise-decaying error.

References

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